ARC-SPEC-2026-EDU-08: Autonomous Agentic Pedagogy, Machine-Taught Systems & Self-Healing Curricula

A Protocol for Closed-Loop Agentic Self-Correction, Automated Laboratory Evolution, and Zero-Human-In-The-Loop Technical Instruction

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Status: Archive-Ready / Path A Academic Extension
Governing Invariants: $C_{\text{ops}} = 0$ | $\tau_{\text{override}} \leq 11.99\text{ms}$ | Pedagogical Drift $(\Delta P) = 0$
Target Baseline: Phase 7 Realization (`v1.3.0-exec`)

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1 Executive Summary & Self-Healing Pedagogy Rationale

ARC-SPEC-2026-EDU-01 through EDU-07 established the complete physical, mathematical, educational, and verification foundation for Deterministic Admissibility Engineering (DAE).

However, as hardware microarchitectures evolve (e.g., new RISC-V extensions, updated eBPF hook sites, next-generation seL4 kernel interfaces), human-written educational curricula slowly decay. Static textbooks become outdated, lab instructions accumulate broken dependencies, and student code templates drift away from bare-metal production realities. In traditional academia, updating courseware requires multi-year departmental reviews, introducing massive operational drag ($C_{\text{ops}} \gg 0$).

This specification formalizes ARC-SPEC-2026-EDU-08: the Autonomic Curriculum Evolution Protocol (ACEP). Using closed-loop autonomous AI agents and formal execution feedback from `substrate-ci.yml`, EDU-08 enables the educational system to self-heal and auto-evolve. When underlying toolchains update or edge microarchitectures shift, autonomous agentic loops update lab harnesses, regenerate test vectors, and refine course materials in real time—maintaining zero pedagogical drift ($\Delta P = 0$) while preserving $C_{\text{ops}} = 0$.

Curriculum Maintenance Comparison

TRADITIONAL CURRICULUM DECAY (Static Textbooks / High Drag):

[Toolchain Shift] $\longrightarrow$ [Labs Break] $\longrightarrow$ [Faculty Committee] $\longrightarrow$ [Manual Revision]

AUTONOMIC SELF-HEALING PEDAGOGY (ARC-SPEC-2026-EDU-08 / Zero Drag):

[Toolchain Shift] $\longrightarrow$ [CI Invariant Failure] $\longrightarrow$ [Agentic Diagnostic Loop]
$\Downarrow$
$\longrightarrow$ [PR Issued & Verified] (Zero Pedagogical Drift)

2 Mathematical Formalization of Pedagogical Drift (ΔP)

To mathematically guarantee that automated curriculum evolution never compromises foundational safety rules or dilutes technical difficulty, we define the Pedagogical Drift Operator (ΔP):

$$\Delta P = \int_{t_0}^{t_1} |\nabla \mathcal{H}_{\text{lab}}(t) - \nabla \mathcal{H}_{\text{spec}}| \, dt \tag{1}$$

Where $\mathcal{H}_{\text{lab}}(t)$ is the entropy and constraint density of the active laboratory code, and $\mathcal{H}_{\text{spec}}$ is the baseline invariant constraint density defined in ARC-SPEC-2026-EDU-01.

$$\text{INVARIANT PRESERVATION: } \Delta P > 0 \vee M_{\text{alloc}} > 0 \vee \text{WCET} > 11.99\text{ms} \implies \text{REFUSAL (POSIX 32)} \tag{2}$$

3 Agentic Self-Healing Protocol Specification (`AGENT-HEAL-01`)

When a breaking change occurs, the `AGENT-HEAL-01` engine executes an automated repair cycle via GitHub Actions.

```
# ARC-SPEC-2026-EDU-08 / Agentic Self-Healing Automation Command
node docs/academic/scripts/agentic_curriculum_repair.js \
  --schema docs/academic/schema/dae_pedagogy_v1.json \
  --error-log harness_audit.log \
  --out-dir docs/academic/harness

# Verify Formal Proofs After Repair
coqc docs/academic/proofs/dae_invariants.v
```

Listing 1: Autonomic Curriculum Self-Healing Workflow Excerpt

Table 1: ACEP Autonomous Feedback Loop Parameters

Feedback Stage	**Automated Substrate Component**	**Safety Preservation Bound**
Regression Detection	`substrate-ci.yml` Toolchain Audit	Triggers upon compilation or latency drift
Diagnostic Engine	`AGENT-HEAL-01` LLM Loop	AST diff analysis against `dae_pedagogy_v1.json`
Verification	Coq Prover (`dae_invariants.v`)	Verifies 0 `Admitted.` statements post-repair
Auto-Merge	GitHub Pull Request Automation	Auto-merges to main branch upon full pass

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SPECIFICATION COMPLETE: ARC-SPEC-2026-EDU-08
DEPLOYMENT POSTURE: ARCHIVED FOR FUTURE PATH A EXTENSION
AUTHORIZATION HASH: A-77-DELTA-SHIELD-LOCKED